Chapter 11 — Hydrocarbons

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Alkanes have formula: (a) C_nH_{2n-2} (b) C_nH_n (c) C_nH_{2n} (d) C_nH_{2n+2}

2. Chlorination of methane needs: (a) Darkness (b) No catalyst (c) Low temperature (d) UV light

3. Homolytic fission gives: (a) Atoms only (b) Molecules only (c) Free radicals (d) Ions

4. Alkenes undergo: (a) Free radical subst. (b) No reaction (c) Nucleophilic subst. (d) Electrophilic addition

5. Markovnikov: H goes to C with: (a) No H (b) More H (c) Less H (d) More Cl

6. Most stable carbocation: (a) Secondary (b) Methyl (c) Primary (d) Tertiary

7. Ethene can be prepared by dehydration of: (a) Ethanol (b) Ethanoic acid (c) Ethane (d) Ethyl chloride only

8. A pi bond forms by: (a) Ionic transfer (b) No overlap (c) Head-on overlap (d) Sideways overlap

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why are alkanes relatively unreactive towards polar reagents?

2. How does UV light initiate chlorination of methane?

3. Why is free-radical substitution a chain reaction?

4. Why are alkenes more reactive than alkanes?

5. How does Markovnikov’s rule predict the major product?

6. Why does ethene decolourise bromine water?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Classify hydrocarbons and explain IUPAC nomenclature of alkanes and cycloalkanes. **(4)**

(b) Explain the free-radical substitution mechanism of alkanes. **(4)**

Q.4 (a) Explain the structure and electrophilic addition reactions of alkenes. **(4)**

(b) Explain carbocation stability and inductive effect. **(4)**

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